

# INTRODUCTION TO AI

**Evolution of AI from Simple Concepts**

# Evolution of AI

- 1950s: Birth of AI
- Machine Learning
- Deep Learning
- Generative AI



# Types of AI

## **NARROW AI** (WEAK AI)



Specialized in  
one task only

Example: Siri,  
Google Translate

## **GENERAL AI** (STRONG AI)



Can think, reason  
& learn like humans

Example:  
Still under research

## **SUPER AI**



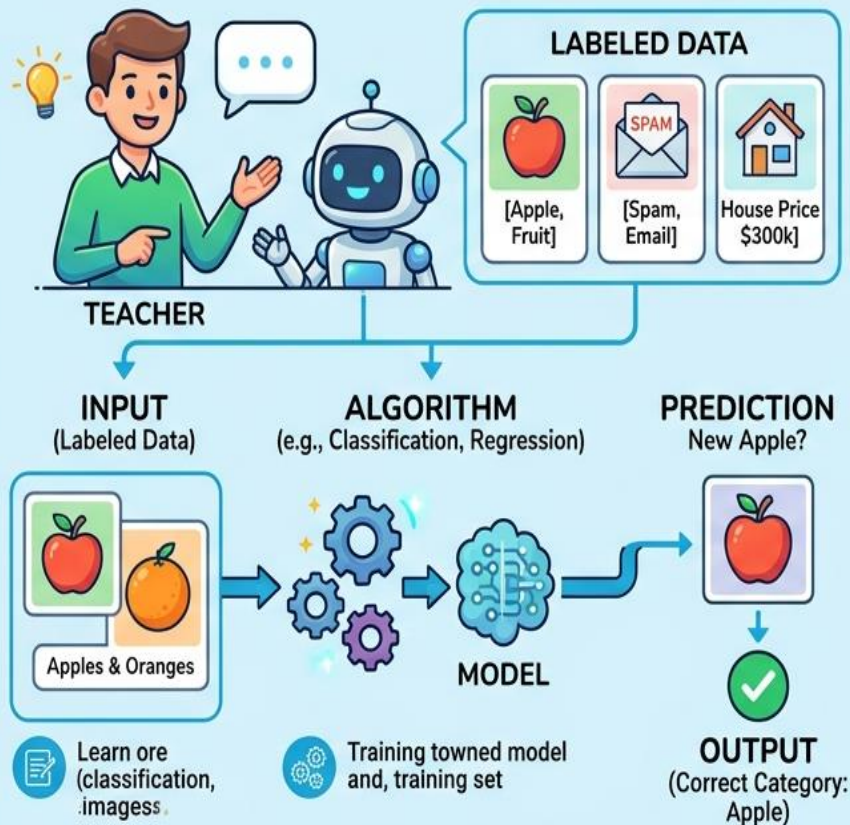
Surpasses human  
intelligence

Example:  
Theoretical / Future

# Supervised vs Unsupervised

## SUPERVISED LEARNING

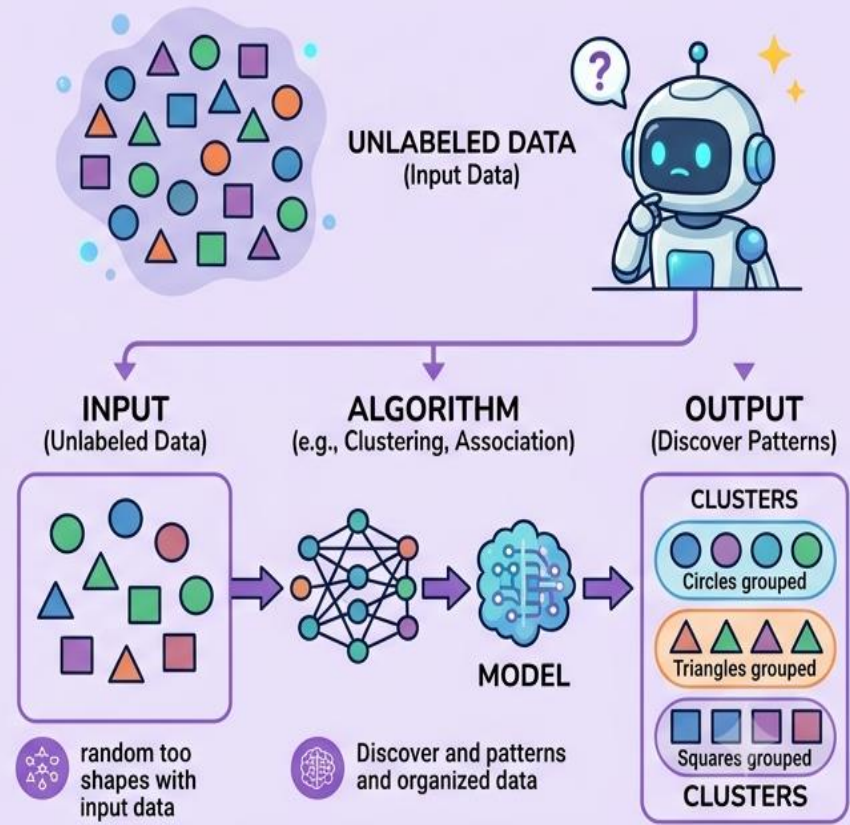
Learning with Guidance (Teacher/Labels)



**Keywords:** Predicted Output, Training Set

## UNSUPERVISED LEARNING

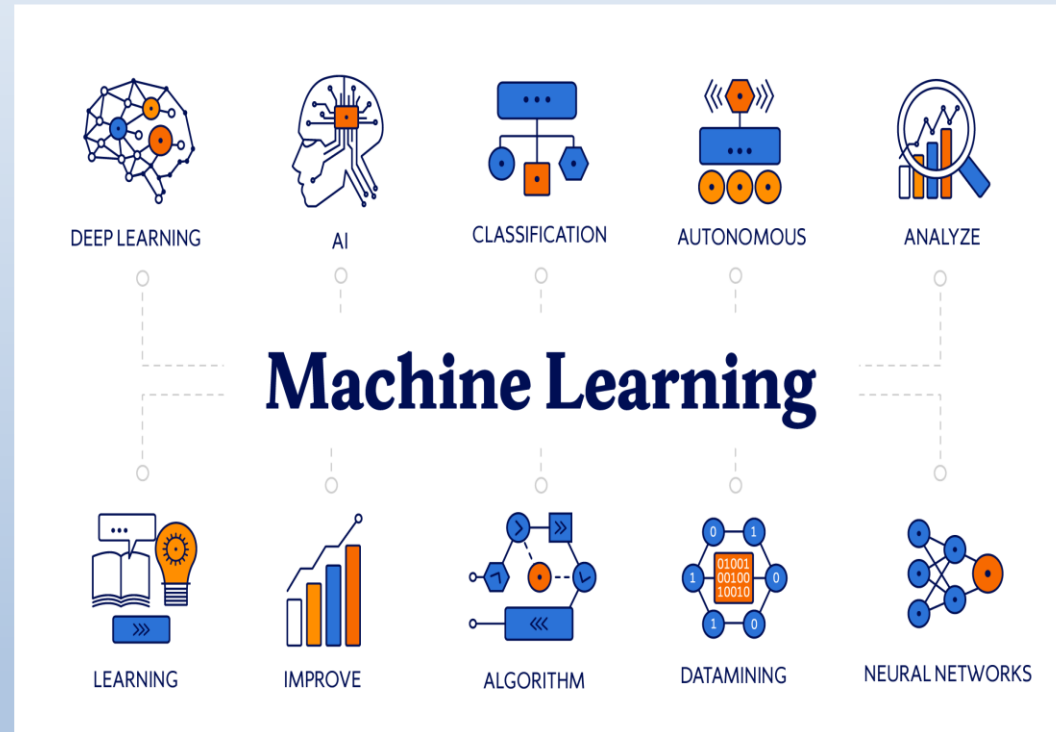
Learning without Guidance (No Labels)



**Keywords:** Patterns & Clusters, Exploration

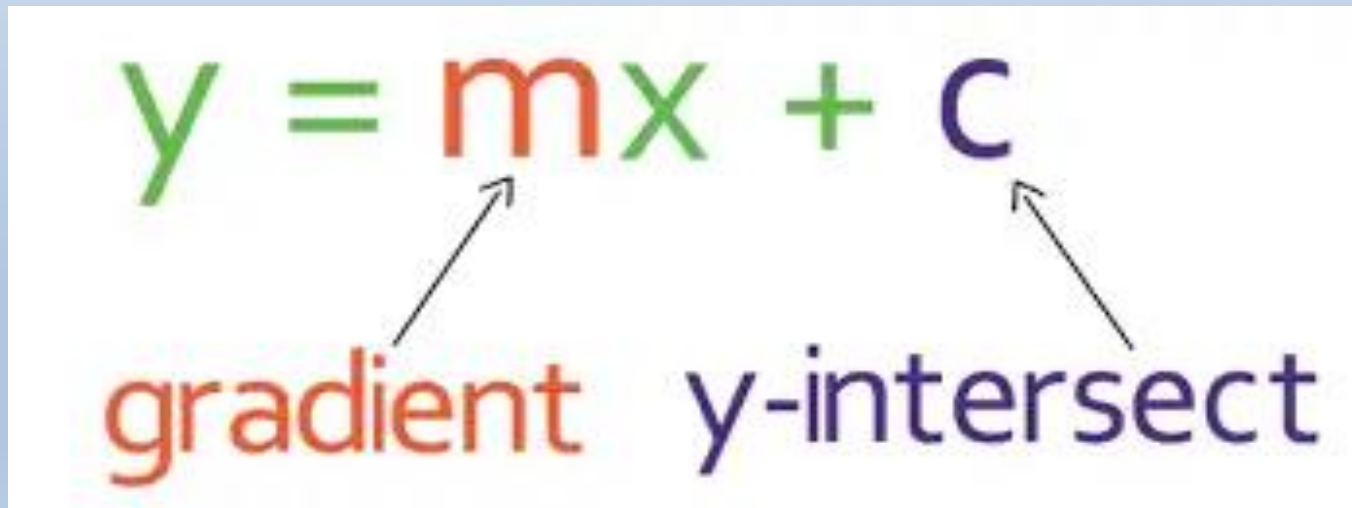
# Machine Learning

- Learns from Data
- Finds Patterns
- Makes Predictions
- Improves Accuracy
- Automates Tasks
- Solves Problems



# Linear Regression

- Equation:  $y = mx + c$
- Predicts continuous values



The diagram shows the linear regression equation  $y = mx + c$  in a large font. Below the equation, the word "gradient" is written in red, and the word "y-intersect" is written in dark blue. Two arrows point from the labels to the corresponding parts of the equation: one arrow points from "gradient" to the variable  $m$ , and another arrow points from "y-intersect" to the variable  $c$ .

$$y = mx + c$$

gradient      y-intersect

# Linear Regression Example

$$X=[1,2,3,4,5] \quad y=[3,5,7,9,11]$$

Numerator:

$$\sum (X - X_{\text{mean}}) \cdot (y - y_{\text{mean}})$$

Denominator:

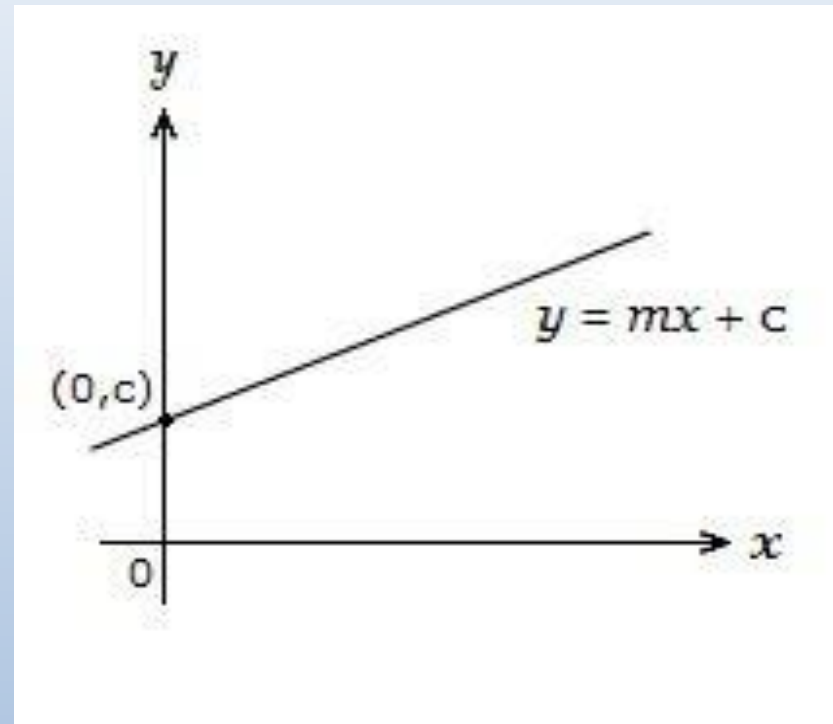
$$\sum (X - X_{\text{mean}})^2$$

**slope (m):**

$$m = \text{Numerator} / \text{Denominator}$$

**intercept (c):**

$$c = y_{\text{mean}} - m \cdot X_{\text{mean}}$$



$$\text{Result: } y=2x+1$$



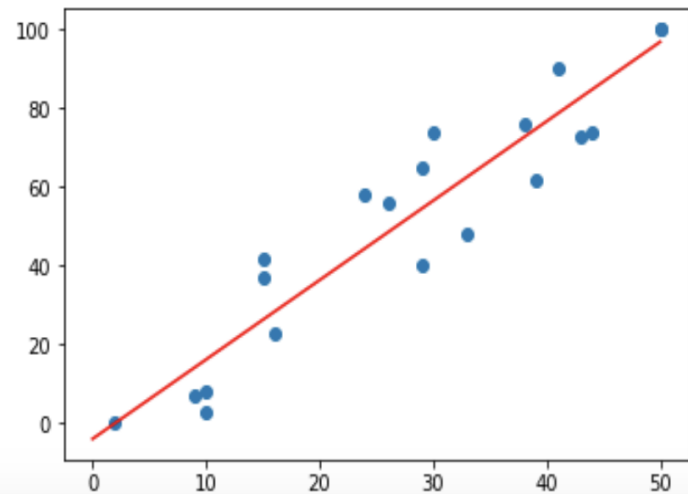
# Python Experiment

- Train Linear Regression
- Predict outputs
- Plot regression line

## Step #6 - Dataviz

```
In [12]: x_lin_reg = range(0, 51)  
y_lin_reg = predict(x_lin_reg)  
plt.scatter(x, y)  
plt.plot(x_lin_reg, y_lin_reg, c = 'r')
```

Out[12]: [<matplotlib.lines.Line2D at 0x1a19bff5550>]



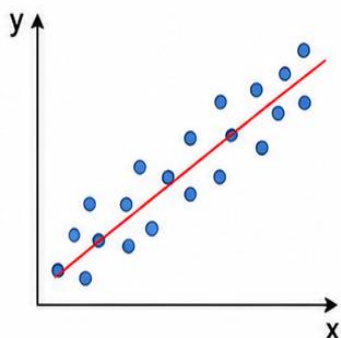


# From Regression to ML

## STEP 1

### LINEAR REGRESSION

Start with a simple linear model to predict outcomes.

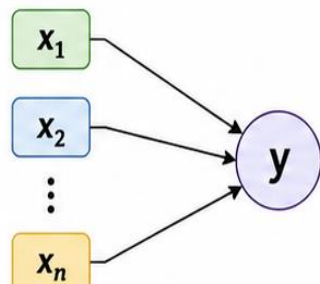


Predicts the relationship between one input (x) and output (y).

## STEP 2

### MULTIPLE LINEAR REGRESSION

Extend to multiple features (inputs) to improve predictions.



#### Equation

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c$$

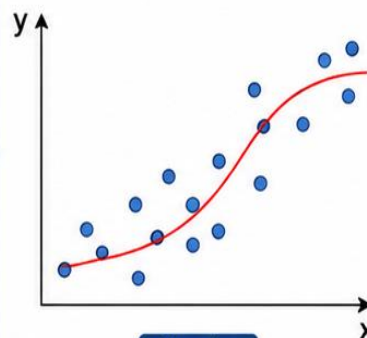


Considers multiple inputs to give better predictions.

## STEP 3

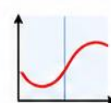
### POLYNOMIAL REGRESSION

Introduce non-linearity by adding polynomial terms.



#### Equation

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$



Captures non-linear patterns in the data.

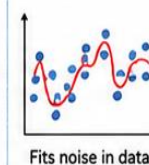
## STEP 4

### REGULARIZATION

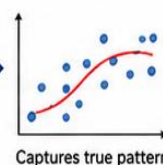
Add techniques like Lasso and Ridge to prevent overfitting.

#### WHY REGULARIZATION?

##### OVERFITTING



##### GOOD GENERALIZATION



#### RIDGE (L2)



Penalizes large coefficients by squaring them.

#### LASSO (L1)



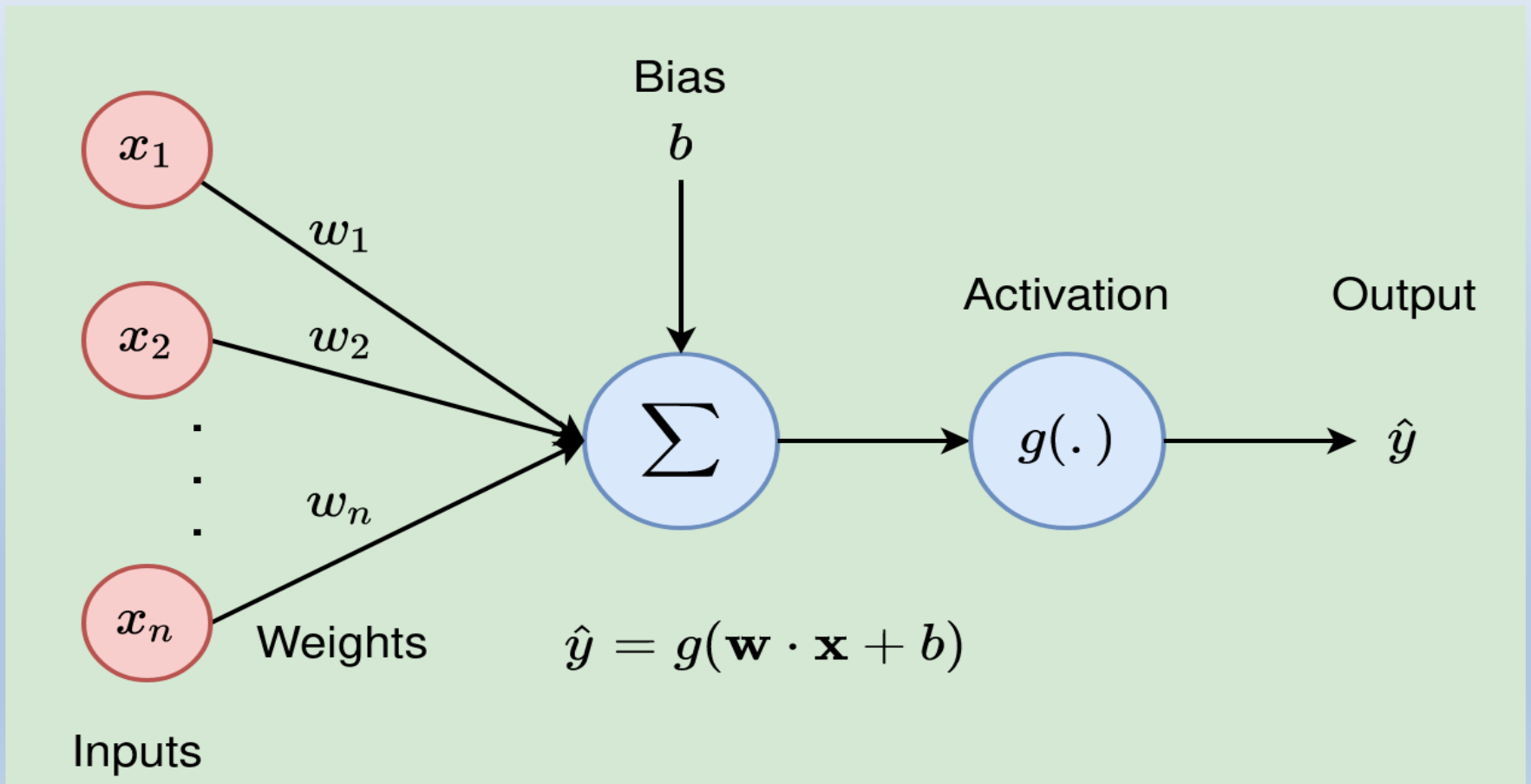
Penalizes large coefficients by absolute value.



Helps prevent overfitting and improves model performance on unseen data.

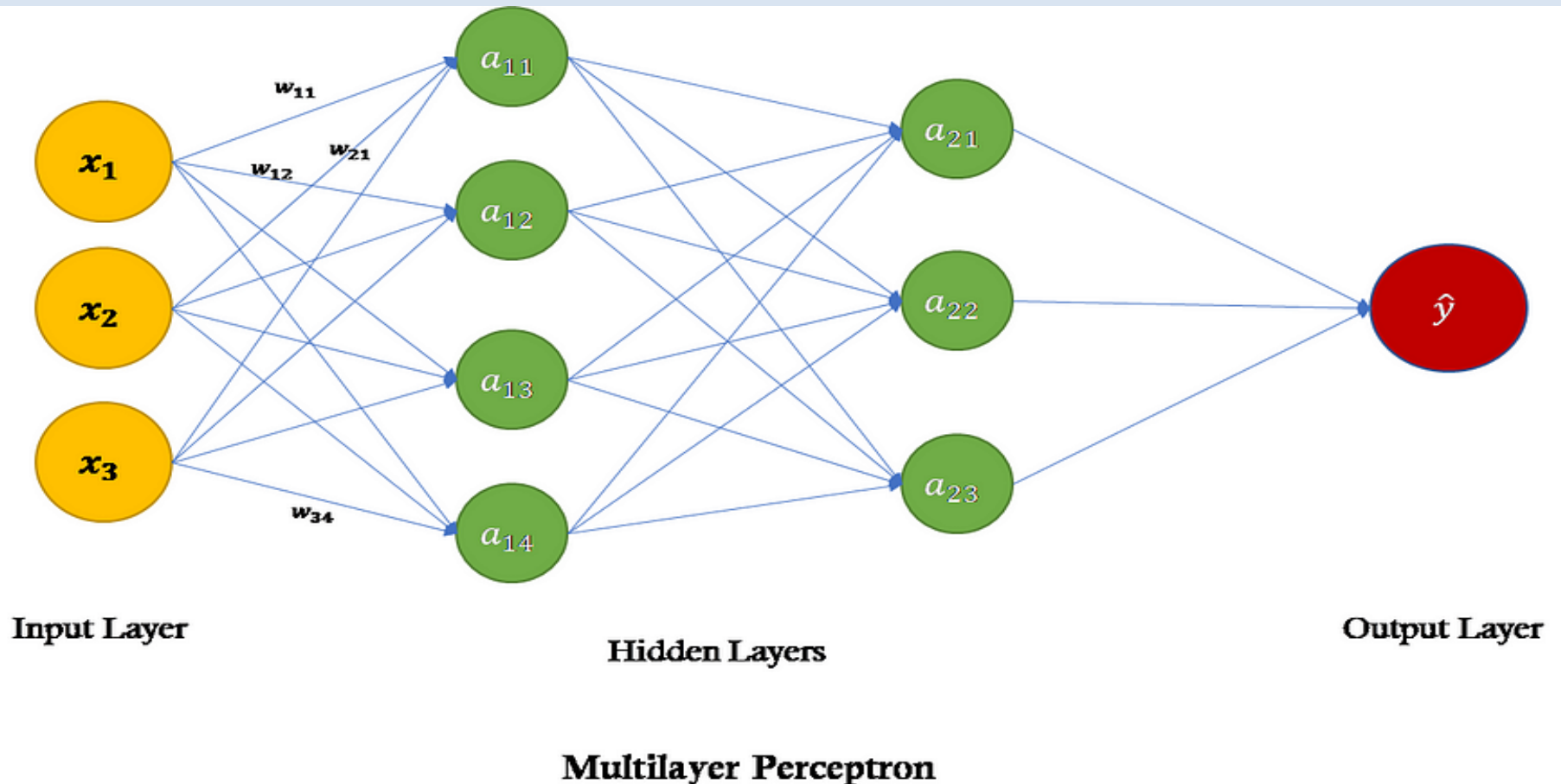
# Artificial Neural Networks

## Single Neuron (Perceptron):

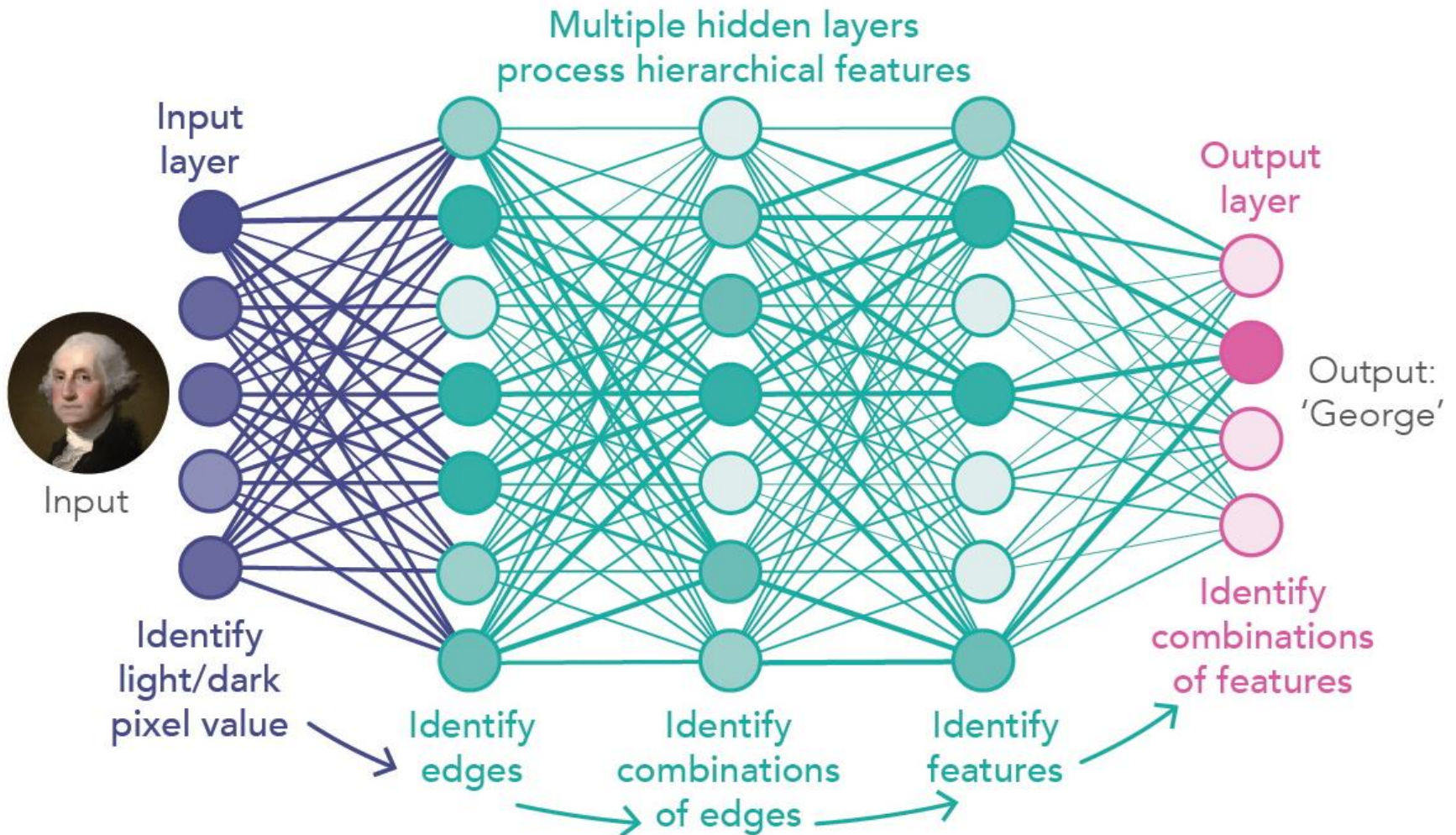


# Artificial Neural Networks

## Multi-Layer Perceptron (MLP):

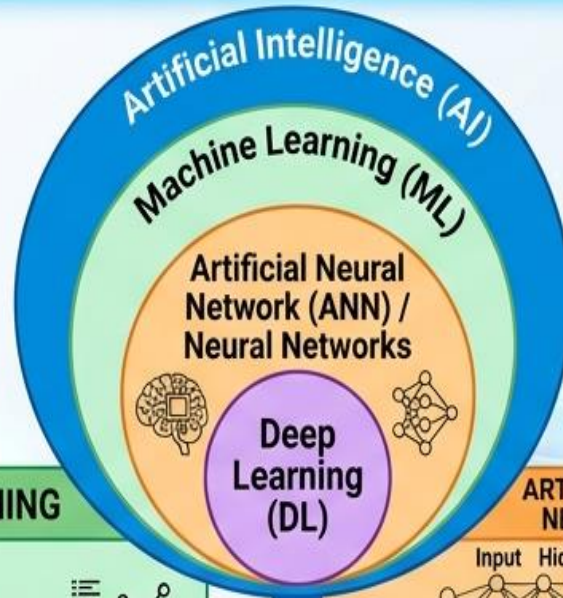


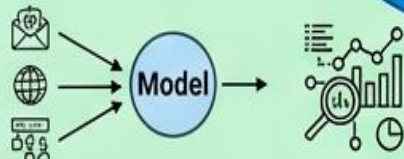
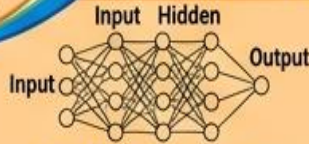
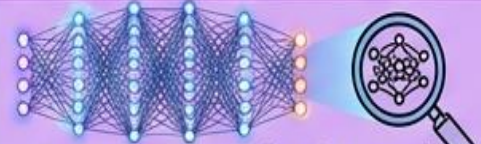
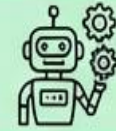
# Deep Learning





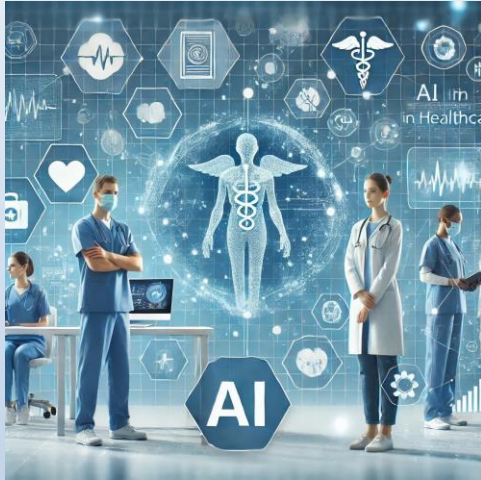
# ML vs ANN vs Deep Learning



| CHARACTERISTICS                  | MACHINE LEARNING                                                                                                    | ARTIFICIAL NEURAL NETWORKS (ANN)                                                    | DEEP LEARNING (DL)                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Visual Structure</b>          |                                    |  |                                                                                                             |
| <b>Data Requirements</b>         | <b>Moderate</b><br>(e.g., thousands of examples)                                                                    | <b>More data</b><br>(e.g., tens of thousands)                                       | <b>Massive</b><br>(e.g., millions, petabytes)                                                              |
| <b>Feature Engineering</b>       | <b>Manual by domain experts</b>  | <b>Some internal feature discovery, but often pre-engineered</b>                    | <b>Automatic</b><br>(Feature extraction learned end-to-end)                                               |
| <b>Interpretability Examples</b> | Spam detection, basic recommendation, fraud detection                                                               | <b>Pattern recognition, function approximation</b>                                  | <b>Image recognition</b><br>(e.g., self-driving car visual), natural language processing, complex gaming  |

Increased Complexity & Data

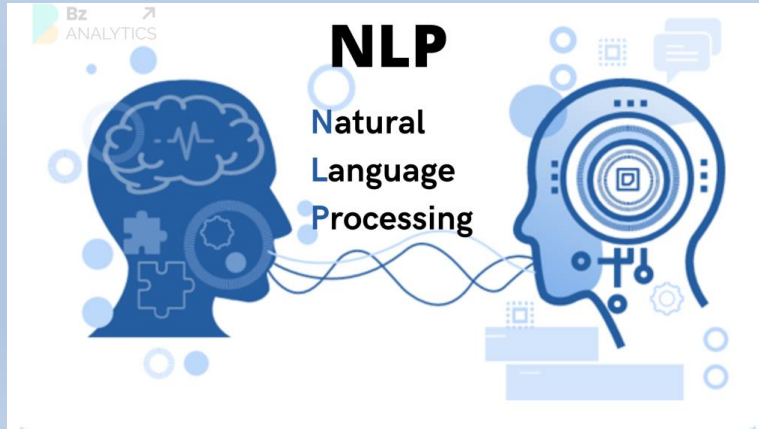
# AI Applications



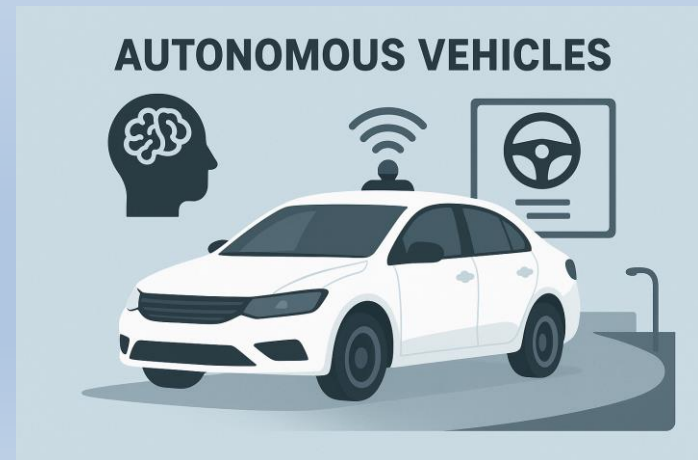
**Healthcare**



**Finance**



**NLP**



**Autonomous Vehicles**

# Conclusion

- Machine Learning helps AI learn from data and make better decisions.
- It is used in many fields like healthcare, finance, education, and transportation.
- Keep exploring Machine Learning to understand how it is shaping the future of AI.